

Comparative Analysis of Retrieval-Augmented Generation Architectures for Financial Question Answering

Uday Jain

Department of Statistics and Machine Learning
Linköping University, Linköping, Sweden 581 83
udaja983@student.liu.se

Abstract

This study evaluates five Retrieval-Augmented Generation (RAG) architectures for financial QA using the FiQA dataset. We compare Naive RAG, Re-Ranking RAG, Hypothetical Document Embeddings (HyDE), Query Expansion RAG, and Multi-Step Iterative RAG across lexical, semantic, and retrieval metrics. Results show Re-Ranking RAG achieves best lexical performance (8% higher ROUGE-L, 177% higher BLEU vs Naive) but with higher variability. Naive RAG provides superior semantic preservation (5% higher cosine similarity) and consistency. Retrieval quality shows limited correlation with generation performance ($r < 0.30$), challenging RAG design assumptions.

1 Introduction

Retrieval-Augmented Generation (RAG) has emerged as a dominant paradigm for knowledge-intensive natural language tasks, combining the strengths of information retrieval and large language models. While the basic RAG architecture is well-established, numerous advanced variants have been proposed to address limitations in retrieval quality, relevance, and answer coherence. However, there remains a lack of systematic comparative evaluation of these architectures, particularly in highly specialised and technical domains such as financial question answering, where precise terminology and factual accuracy are critical.

Financial domain QA presents unique challenges (1) Terminology specificity: Domain-specific jargon requires precise retrieval (2) Factual precision: Incorrect financial information can have serious consequences (3) Regulatory compliance: Answers must align with established financial principles (4) Multi-faceted queries: Often combine quantitative and qualitative aspects. This study addresses this gap by implementing and evaluating five RAG architectures.

Using the FiQA financial dataset (Maia et al., 2018), we conduct a comprehensive evaluation across **12 complementary metrics** spanning lexical, semantic, and retrieval dimensions to answer three research questions:

1. How do advanced RAG architectures compare to the naive baseline in financial QA, particularly regarding domain-specific terminology handling?
2. What is the relationship between retrieval quality metrics and final generation quality across different architectures?
3. Which evaluation metrics provide the most discriminative power for architecture comparison in financial domains?

Our contributions include: (1) Implementation framework: Open-source modular framework for RAG architecture experimentation (2) Empirical evidence: Comprehensive benchmarking providing architecture selection guidance for financial QA (3) Metric insights: Analysis of which evaluation metrics best capture architectural differences in specialised domains (4) Domain-specific findings: Identification of architecture behaviours specific to financial terminology and precision requirements

2 Theoretical Background

2.1 Retrieval-Augmented Generation

The RAG framework (Lewis et al., 2020) combines a retrieval component $R(q)$ that retrieves relevant documents D for query q , and a generation component $G(q, D)$ that produces an answer conditioned on both the query and retrieved documents: $Answer = G(q, R(q))$

This architecture addresses two key limitations of pure LLMs: (1) reliance on parametric knowledge that may be outdated or incomplete, and (2) hallucination when generating facts not in training

data. By grounding generation in retrieved evidence, RAG improves factuality and allows access to current information.

2.2 Advanced RAG Architectures

Re-Ranking RAG employs a two-stage retrieval process: initial retrieval using dense embeddings followed by re-ranking with a cross-encoder model (Nogueira et al., 2019). This addresses the *double penalty* problem where semantically relevant but lexically different documents are missed by single-stage retrieval. The final relevance score combines both stages: $s_{\text{final}} = \alpha \cdot s_{\text{dense}} + (1 - \alpha) \cdot s_{\text{cross}}$

where s_{dense} represents cosine similarity from dense retrieval and s_{cross} represents relevance probability from the cross-encoder.

Hypothetical Document Embeddings (HyDE) (Gao et al., 2023) generates a hypothetical answer to the query using the LLM, then uses this as the retrieval query. This approach bridges the *lexical gap* between user queries and relevant documents by mapping from query space to answer space: $q_{\text{hyde}} = \text{LLM}(q_{\text{original}})$

The intuition is that the hypothetical answer shares more lexical and semantic similarity with relevant documents than the original query.

Query Expansion RAG generates multiple query variations to address vocabulary mismatch and increase retrieval recall (Zhuang and Zuccon, 2023). Given original query q , the system generates n expansions $\{q_1, q_2, \dots, q_n\}$: $R_{\text{expanded}}(q) = \bigcup_{i=1}^n R(q_i)$ The union of retrieved documents from all expansions provides broader coverage, particularly valuable for technical domains where terminology may vary.

Multi-Step Iterative RAG employs an iterative retrieval process where intermediate answers inform subsequent retrieval steps (Jiang et al., 2023), mimicking human reasoning in complex QA through query refinement based on intermediate answers.

2.3 Evaluation Framework

We employ a multi-faceted evaluation approach addressing different quality dimensions:

- **Lexical Metrics:** ROUGE-1/2/L/LSum (Lin, 2004) measure n-gram overlap. ROUGE-1 (unigrams) captures content coverage,

ROUGE-2 (bigrams) assesses phrase structure, while ROUGE-L (Longest Common Subsequence) evaluates sentence-level similarity.

- **Semantic Metrics:** BERTScore (Zhang et al., 2019) computes similarity using contextual BERT embeddings, with precision (how much of generated content is relevant), recall (how much of reference content is covered), and F1 (harmonic mean).
- **Fluency Metrics:** BLEU score (Papineni et al., 2002) evaluates word choice and sentence structure quality, particularly important for domain-specific terminology usage.
- **Embedding Similarity:** Cosine similarity between answer embeddings provides a holistic semantic similarity measure complementary to BERTScore.
- **Retrieval Quality:** Average retrieval score indicates retrieval confidence, while retrieval similarity to ground truth context measures relevance accuracy.

This comprehensive framework allows us to assess not only *what* is said (lexical/semantic content) but also *how well* it is said (fluency) and *how* it was derived (retrieval quality).

3 Data and Methodology

3.1 Dataset

We use the FiQA dataset (Maia et al., 2018), a financial question answering corpus via hugging face containing 57,638 documents to search for retrieval followed by 30 question-answer pairs with optimal retrieved contexts.

3.2 Implementation Details

All architectures share common components: (1) Embedding Model(all-MiniLM-L6-v2 384 dimensions) (2) Vector Database (Qdrant; cosine similarity indexing) (3) LLM (Local 2.7B param microsoft/phi-2 endpoint) (4) Reranker(cross-encoder/ms-marco-MiniLM-L-6-v2)

3.3 Parameter Optimization

Each architecture was optimized through grid search. Naive RAG($k \in \{1, 2, 3, 4, 5\}$, optimal at $k=4$. See 3), Re-Ranking ($k=4$, $\text{rerank}_k = 20, \alpha=0.7$. See 4, 5), HyDE($k=4$, temperature=0.3. See 6), Query Expansion($k=2$, numExpansions=4.

See7,8) and Multi-Step(k=2, kStep=3, nSteps=2. See9,10,11).

3.4 Evaluation Protocol

Each query was processed through all five architectures, with answers evaluated against ground truth using our 12-metric framework. Given the exploratory nature of this study and the computational constraints of running statistical tests across 12 metrics $\times$ 5 architectures $\times$ 30 queries, we focus on descriptive statistics.

4 Results

4.1 Overall Performance Comparison

Figure 1 presents the mean scores with standard error bars for each architecture across all metrics. The visual comparison reveals clear architectural trade-offs: Re-Ranking RAG excels in lexical metrics but with higher variability, while Naive RAG provides more consistent semantic preservation.

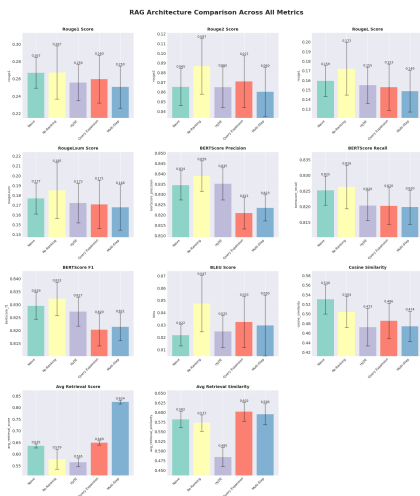


Figure 1: Comparative analysis of RAG architectures across evaluation metrics. Error bars represent standard errors

4.2 Correlation Analysis Revealing Metric Relationships

Correlation analysis (Figure 2) reveals critical RAG evaluation insights:

Metric Redundancy Near-perfect ROUGE inter-correlations (ROUGE-L vs LSum: $r=0.98$) make reporting multiple variants redundant. Strong BLEU-ROUGE correlation ($r=0.91$) indicates overlapping constructs.

Retrieval-Generation Independence Weak correlations ($r<0.30$) support separate evaluation. Retrieval similarity correlates moderately with semantic metrics ($r=0.49$) but negligibly with lexical metrics.

Practical Recommendations We recommend: ROUGE-L (lexical), Cosine + BERTScore F1 (semantic), BLEU (fluency), and retrieval similarity only. This reduces 12 to 5 metrics while maintaining coverage.

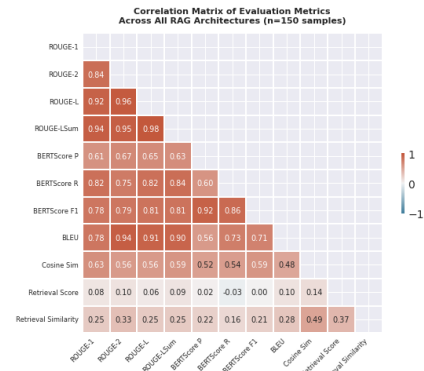


Figure 2: Correlation matrix between evaluation metrics across all architectures (n=150 samples)

4.3 Key Observations

Lexical Performance with Variability Re-Ranking RAG outperforms all other architectures across all ROUGE metrics (ROUGE-L: 0.172 vs 0.159 for Naive, **8% relative improvement**). However, this comes with substantially higher standard errors (ROUGE-L SE: 0.027 vs 0.016 for Naive), indicating greater response inconsistency. This suggests that while cross-encoder re-ranking can significantly improve lexical alignment, its effectiveness varies across queries.

Semantic Preservation in Simpler Architectures Despite lower lexical scores, Naive RAG maintains superior cosine similarity (0.530 vs 0.504 for Re-Ranking, **5% higher**) and competitive BERTScore metrics (F1: 0.829 vs 0.832 for Re-Ranking). Coupled with the lowest standard errors across metrics, this indicates that simpler architectures provide more reliable semantic preservation, albeit with less lexical precision.

Fluency as Key Discriminator BLEU scores show the largest architectural differences, with Re-Ranking outperforming Naive by **177%** (0.047 vs 0.021). This substantial gap suggests that re-ranking significantly improves word choice and

sentence structure fluency, making BLEU a particularly sensitive metric for architecture comparison in financial QA.

Retrieval-Generation Decoupling Despite having the lowest average retrieval score (0.578 vs 0.635 for Naive, **9% lower**), Re-Ranking produces the best answers. This counterintuitive finding suggests that retrieval confidence metrics may not correlate with generation quality, possibly because high-confidence retrievals lack the diversity needed for comprehensive answers and because of smaller LLM.

HyDE’s Domain-Specific Underperformance : HyDE shows the poorest retrieval similarity (0.485 vs 0.582 for Naive, **17% lower**), suggesting that hypothetical document generation struggles with financial domain precision. This performance appears highly dependent on the LLM’s ability to generate accurate hypothetical financial content.

Query Expansion’s Limited Impact : Despite generating multiple query variations, Query Expansion showed limited improvements over Naive RAG (ROUGE-L: 0.153 vs 0.159, **4% lower**) and reduced cosine similarity (0.486 vs 0.530, **9% lower**). This suggests that query expansion alone, without corresponding refinement mechanisms, provides limited benefit.

Multi-Step’s Retrieval-Quality Mismatch : Multi-Step achieves the highest retrieval similarity (0.596) and exceptionally high average retrieval score (0.824), but this doesn’t translate to generation quality (lowest ROUGE scores: 0.149 vs 0.172 for Re-Ranking). The high retrieval scores with similar documents suggest limited diversity and the iterative process appears constrained by LLM performance bottlenecks.

5 Discussion

5.1 Interpretation of Architectural Trade-offs

The superior lexical performance of Re-Ranking RAG can be attributed to its dual-stage retrieval process supports the theoretical advantage of cross-encoder refinement for domain-specific retrieval. In financial contexts where precise terminology matters (e.g., distinguishing between "401(k)" retirement plans and general investment options), the semantic understanding of cross-encoders appears particularly valuable. However, the observed variability (standard errors 72% higher than Naive

RAG for ROUGE metrics) suggests this advantage is query-dependent or inherently cross-encoders dependent, recommending use of domain specific trained cross-encoders where domain specific relevance takes precedence over semantic similarity.

The strong semantic performance of Naive RAG despite poor BLEU scores suggests a trade-off: simpler architectures may preserve semantic meaning but struggle with fluency and structure. This aligns with findings by [Lewis et al. \(2020\)](#) that retrieval quality doesn’t linearly translate to generation quality. Comparable overall performance coupled with a strong consistency also presents a compelling case for simpler architectures in production systems unless custom domain trained embeddings are used. The minimal performance gap in BERTScore metrics (0.003 difference in F1) coupled with significantly lower standard errors suggests that for many financial QA applications, the additional complexity of advanced architectures may not justify their computational cost unless coupled with custom trained models on domain specific data.

5.2 The BLEU Metric as Architectural Discriminator

The dramatic BLEU score differences (177% improvement for Re-Ranking) highlight this metric’s sensitivity to architectural improvements in financial QA. Unlike general domains where BLEU may penalize legitimate paraphrasing, in financial contexts where precise terminology is crucial, BLEU effectively captures architecture’s ability to generate fluent, domain-appropriate language. This makes BLEU particularly valuable for architecture comparison in technical domains.

5.3 Retrieval Quality Misalignment

The inverse relationship between retrieval confidence metrics and generation quality challenges conventional assumptions about RAG system design. Our findings suggest that:

1. High retrieval similarity scores may indicate *overfitting* to a narrow set of documents, limiting answer comprehensiveness
2. Moderate-confidence retrievals with higher diversity better support the LLM’s generation capabilities
3. Current retrieval metrics may need augmentation with diversity measures to better predict generation quality

5.4 Limitations and Methodological Considerations

Study limitations contextualize findings:

1. **Single-domain evaluation:** Financial QA characteristics may not generalize
2. **LLM constraints:** Single 2.7B parameter model limits generalizability
3. **Statistical analysis:** Descriptive patterns without formal hypothesis testing
4. **Computational scope:** 30 queries may represent limited variability

Despite limitations, performance magnitude and consistent patterns provide meaningful architecture selection insights.

5.5 Related Work in RAG Evaluation

Systematic architectural RAG comparisons remain limited. Our HyDE results align with Gao et al. (2023)'s variable performance findings, though we use a technical domain dataset versus general ones. Re-Ranking success supports Nogueira et al. (2019)'s cross-encoder effectiveness. Zhuang and Zuccon (2023) demonstrated query expansion in web search but not specifically for RAG. Jiang et al. (2023) proposed iterative retrieval without comprehensive benchmarking. Our study addresses this gap through systematic five-architecture comparison in financial QA with domain-specific multi-metric evaluation.

5.6 Qualitative Analysis

Manual inspection of samples revealed Re-Ranking's precise terminology and source citation, Naive's verbosity with semantic correctness, HyDE's occasional hallucinations, Query Expansion's broader concept coverage, and Multi-Step's document redundancy due to LLM bottlenecks.

5.7 Reproducibility

Our complete implementation is available at: https://github.com/SpikeStriker/732A81_text_mining_research_project/tree/main/EvaluatingRagArchitecture.

6 Conclusion

This study provides empirical evidence for RAG architecture selection in financial question answering through comprehensive evaluation across five architectures and twelve metrics. Our key findings reveal important trade-offs and considerations:

1. **Performance-Variability Trade-off:** Re-Ranking RAG offers superior lexical performance and fluency but with higher response variability, while Naive RAG provides more consistent semantic preservation with competitive performance.
2. **Metric Sensitivities:** Different metrics reveal distinct architectural strengths—BLEU best captures fluency improvements, ROUGE identifies lexical precision, while cosine similarity highlights semantic preservation.
3. **Retrieval-Generation Decoupling:** Retrieval quality metrics show limited correlation with generation quality, suggesting that current retrieval evaluation may not adequately predict final answer quality.
4. **Domain-Specific Architecture Suitability:** In financial QA where precise terminology is critical, Re-Ranking's cross-encoder refinement provides clear advantages, though simpler architectures remain competitive for semantic-focused applications.

These findings have practical implications for RAG system design in specialized domains: architects must balance the precision benefits of advanced techniques against their computational costs and variability. The choice ultimately depends on application priorities—precision vs consistency, fluency vs semantic preservation, and performance vs cost.

6.1 Future Research Directions

- **Statistical validation:** Larger-scale studies with formal hypothesis testing to confirm observed patterns
- **LLM impact analysis:** Systematic investigation of how different LLMs (size, architecture, training) interact with RAG architectures
- **Dynamic architecture selection:** Developing query-based heuristics for selecting optimal architectures per query type
- **Cross-domain evaluation:** Testing architectural patterns across multiple specialized domains (legal, medical, technical)

Acknowledgments

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A RAG Parameter Optimisations

A.1 Naive RAG

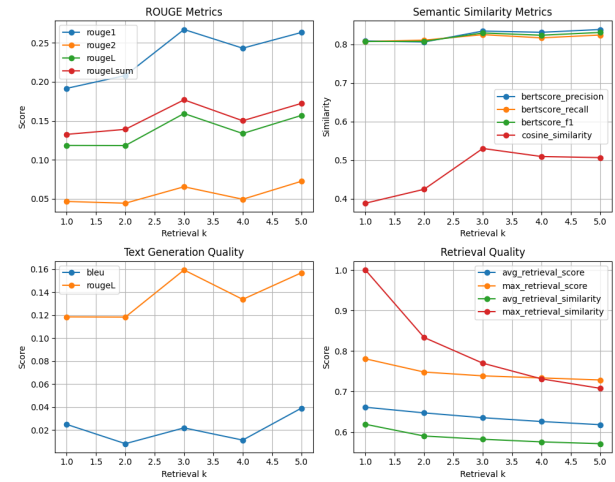


Figure 3: Optimizing for k Retrieval

A.2 Re-Ranking RAG

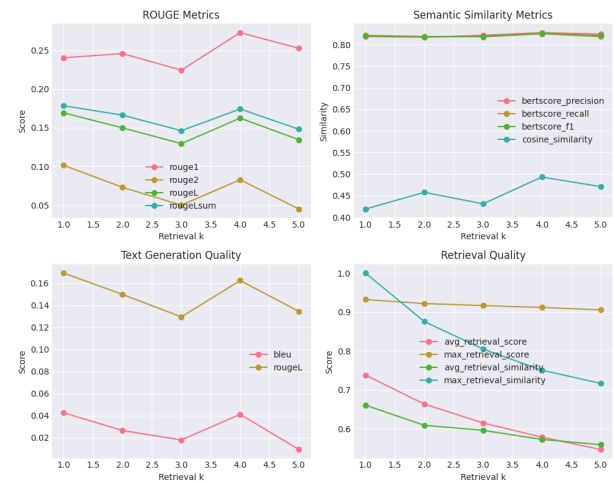


Figure 4: Optimizing for k Retrieval

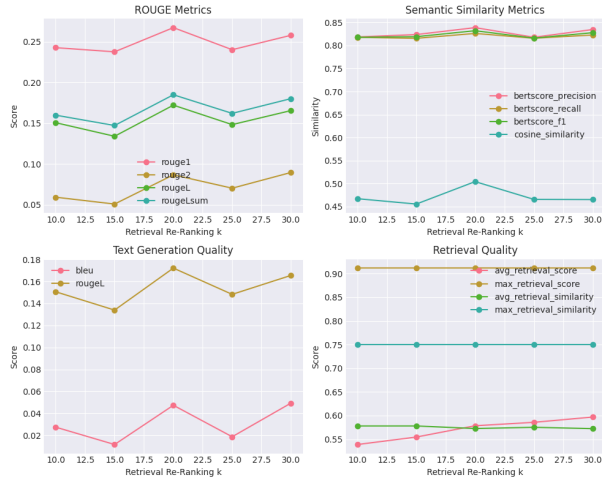


Figure 5: Optimizing for k Re-Ranking Retrieval

A.4 Query Expansion RAG

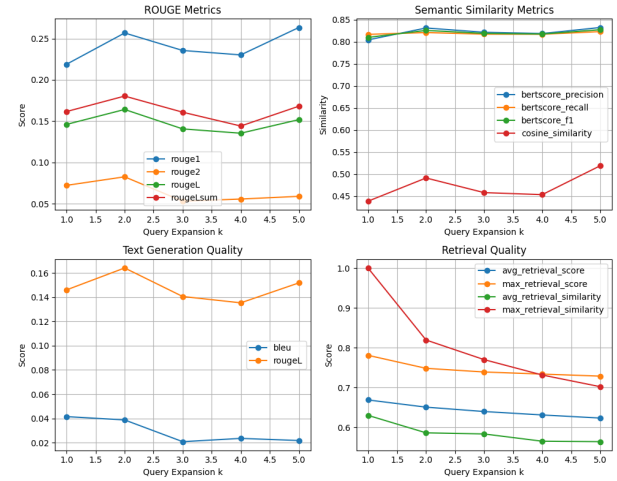


Figure 7: Optimizing for k Retrieval

A.3 Hypothetical Document Extraction (HyDE) RAG

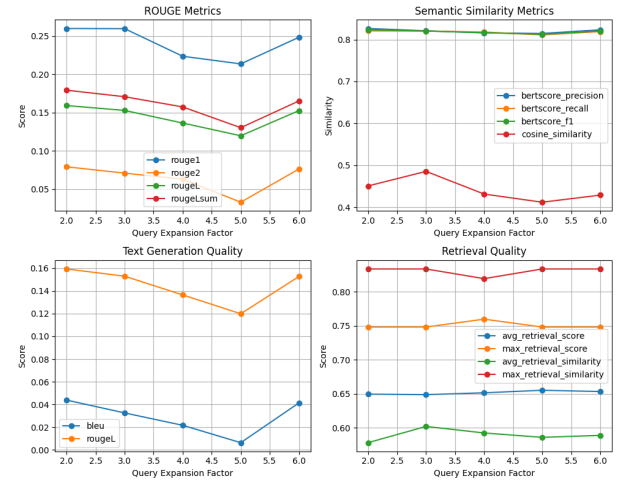


Figure 8: Optimizing for k Query Expansions

A.5 Iterative Multi-step Retrieval RAG

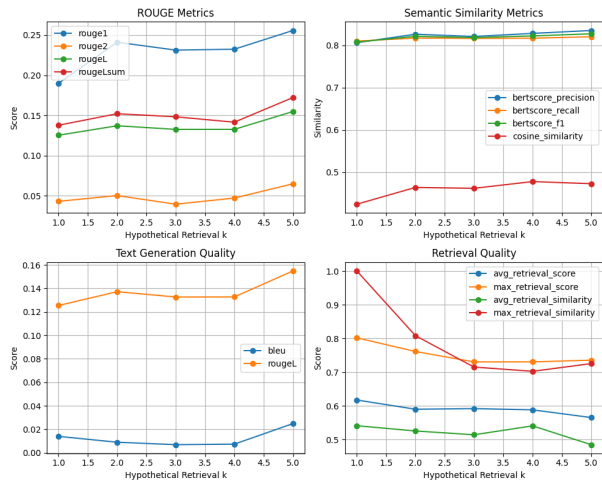


Figure 6: Optimizing for k Retrieval

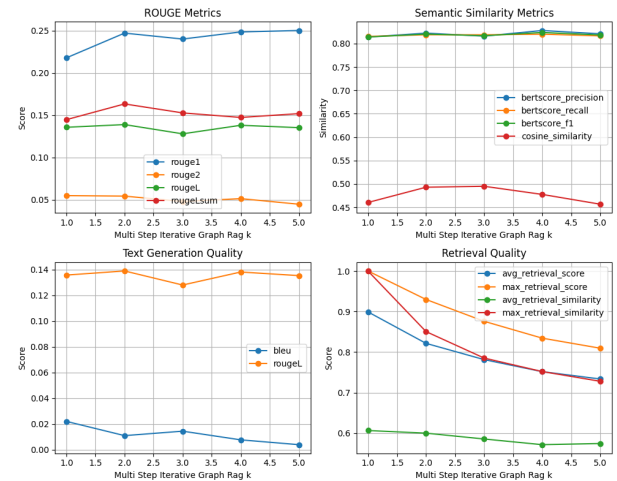


Figure 9: Optimizing for k Retrieval

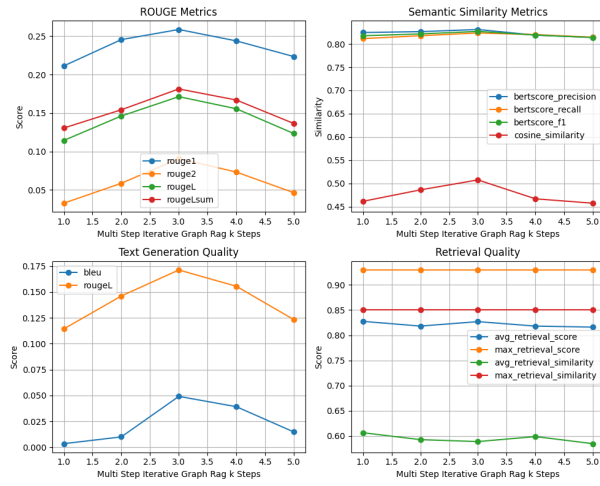


Figure 10: Optimizing for k iterations

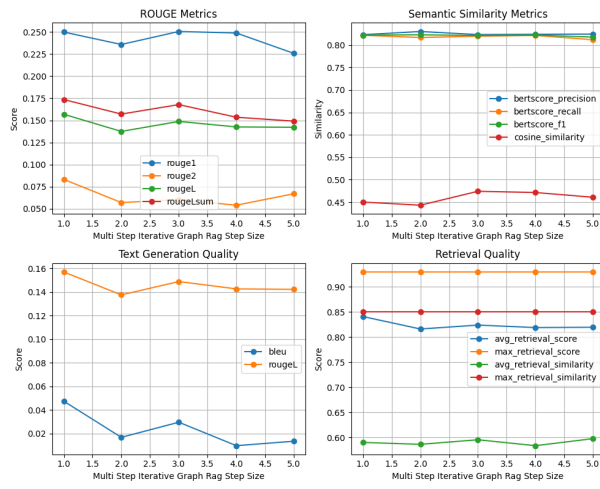


Figure 11: Optimizing for k Retrievals in each step